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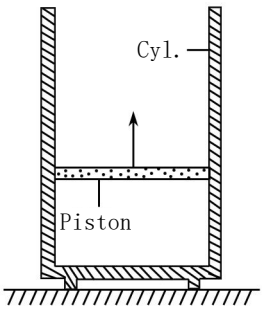
27th National High School Physics Competition: Final Problems

I. (25 points) Fill in the blanks.

1. A uniform thin rubber ring has mass M, stiffness k and natural radius R. Thrown aloft, it stays circular in steady motion. Ignore gravity. It translates and rotates uniformly at ω about its central normal axis. Its radius is _____.

2. A pigeon's whistle has frequency f. Its top flight speed is u. The Doppler effect makes observed frequencies range from _____ to _____. (Sound travels at V.)

3. A vertical insulated cylinder has mass M and inner area A, with a fixed amount of ideal gas at T_0, sealed by a piston. The piston is insulated too. Neglect its mass and friction. Atmospheric pressure is p_0; gravity is g. Pull the piston up slowly. At the opening, the cylinder just leaves the ground. In a slow adiabatic change pV^γ is fixed; p is pressure, V is volume. γ is constant. At the opening, the gas temperature is _____.



4. Use two significant digits. Electron microscopes replace light with electrons to image objects. To resolve $1.0\times10^{-10}\,\mathrm{m}$ (atomic size), the wavelength cannot be larger. Thus the minimum electron speed is _____. To resolve $1.0\times10^{-12}\,\mathrm{m}$ structures, the minimum speed is _____. The required accelerating voltage is _____. The electron has rest mass $m_e=9.1\times10^{-31}\,\mathrm{kg}$ and charge $e=-1.6\times10^{-19}\,\mathrm{C}$. Planck's constant is $h=6.7\times10^{-34}\,\mathrm{J\cdot s}$; the speed of light is $c=3.0\times10^{8}\,\mathrm{m\cdot s^{-1}}$.

II. (20 points) As shown, a point-mass block slides from a platform down a slope onto a long horizontal belt at speed v that carries it far away. The incline's height is $h=2.0\,\mathrm{m}$, horizontal run $L=4.0\,\mathrm{m}$, belt width $d=2.0\,\mathrm{m}$, speed $v=3.0\,\mathrm{m/s}$. Friction is $\mu_1=0.30$. The block starts from rest at the top; while sliding, its velocity is normal to belt motion. A tiny circular arc joins the incline and belt, preserving its speed at the junction. The gravitational acceleration is $g=10\,\mathrm{m/s^2}$.

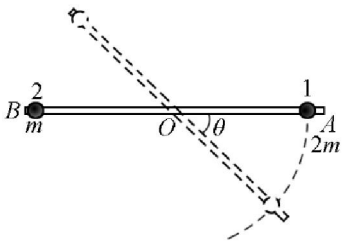
1. What is the minimum block-belt friction coefficient μ_2 to keep the block from sliding off the belt?

2. A DC motor with a speed regulator drives the belt. The connected supply has emf

$E = 200\text{V}$, with zero resistance. Motor resistance is $R = 10\Omega$; its no-load current is $I_0 = 2.0\text{A}$. For a steady mean cargo flow (mass conveyed per unit time) of $\eta = \frac{640}{9}\text{kg/s}$, find the mean motor current. Apart from cargo-belt friction and Joule heat, assume that all other energy losses are independent of the cargo load on the belt.

III. (20 points) A light thin rigid rod pivots on a smooth horizontal axle through O, freely in a vertical plane. Balls 1 and 2, threaded on it, are point masses $2m$ and m. Static friction has coefficient $\mu = \frac{5\sqrt{3}}{6}$. The rod is horizontal at rest; balls 1 and 2 are next to ends A and B, respectively. Release the system from rest. Find:

1. the exit position of ball 1, as the rod's angle to the horizontal at ball 1's exit;
2. the exit position of ball 2, as the rod's angle to the horizontal at ball 2's exit.



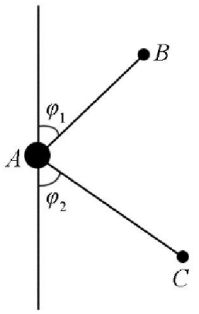
IV. For particles A, B, C, A is far heavier than B, C; B and C have equal masses. A attracts B, A attracts C, while B repels C. They move under these mutual forces. With suitable forces, the following motion is possible:

The motion has these properties:

A, B, C keep fixed relative positions and rotate about an axis in their plane, uniformly. Since A is far heavier than B and C, take the axis fixed through A.

The angles of AB, AC to the axis, φ_1, φ_2, differ; $0<\varphi_1<\frac{\pi}{2}$, $0<\varphi_2<\frac{\pi}{2}$.

The attractions are $f_{AB}=k|AB|^a$ for AB and, for AC, $f_a=k|AC|^a$; $|AB|$, $|AC|$ are the A–B, A–C distances. k is a proportionality constant. Ignore gravity. For which a can this motion occur?



V. (15 points) A huge iceberg breaks off an Antarctic ice shelf, drifting near a city. Use the iceberg as follows. (a) Seal ambient air in a variable-volume vessel. Cool the air at constant pressure to the iceberg temperature by contact. (b) Remove the vessel from the iceberg. At fixed volume, let its cold air absorb ambient heat to ambient temperature. (c) Eject air at fixed volume to run a generator. Repeat until all ice melts. The surroundings are at $T_a=293\text{K}$, the ice at $T_1=273\text{K}$. The available ice mass is $m=1.0\times10^{11}\,\text{kg}$. To estimate the electrical energy, the designer makes these assumptions and uses the following data:

1. Air is an ideal gas.

2. Ice has latent heat of fusion $L = 3.34 \times 10^5\ \mathrm{J/kg}$. Meltwater at T_1 is not reused.

3. At pressure p and volume V, air has internal energy $U = 2.5pV$.

4. Good heat exchange keeps the air in the vessel isothermal during ejection.

5. Treat ejection as steps, each expelling volume u, with u far below the vessel volume. In each step, nozzle gas accelerates under the pressure difference and gains ΔE in kinetic energy and exits. Ignore its volume change while accelerating. Assume vessel pressure stays uniform during ejection; the outside pressure is atmospheric.

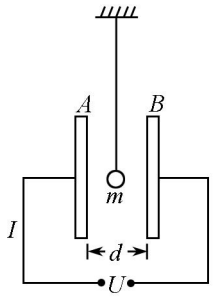
6. The electrical energy obtainable is 45% of the sum of ΔE.

7. If $|x| \ll 1$, then $\ln(1+x) \approx x$.

Using these assumptions, find the electrical energy obtainable from the iceberg.

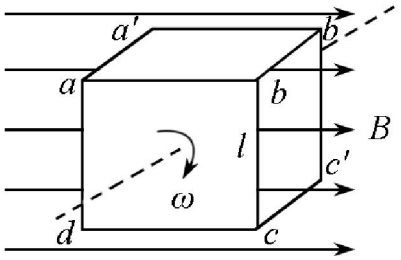
VI. (15 points) Large vertical parallel metal plates A, B are d apart, at constant voltage U. A metal-coated table-tennis ball of mass m hangs between them. Nudge it towards one plate. It rebounds, hits the other plate, and repeats. Collisions are inelastic, with restitution coefficient e. Contact is very brief, during which electrostatic equilibrium is reached. The ball's charge q relates to plate voltage by $|q| = C_0 U$, where C_0 is constant. Assume the ball radius is far below plate spacing d, and its charge does not affect plate charge distribution. A long enough string makes the motion nearly horizontal and straight. Neglect initial kinetic energy at the first collision and air resistance. For the setup shown, find:

1. the maximum kinetic energy attainable by the moving ball;
2. the mean current in the external circuit after a long time.

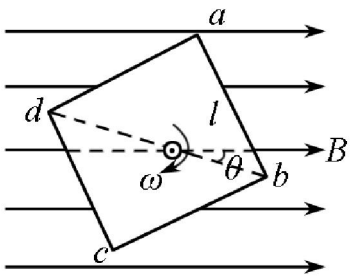


VII. (20 points) In Fig. (a), 12 uniform wire rods form a rigid cube of edge l. Each rod has resistance R. It rotates uniformly about a central axis normal to face $abcd$ at angular speed ω, in a uniform magnetic field of magnitude B normal to that axis. Neglect self-inductance. In Fig. (b), diagonal bd makes angle θ with the field. Find:

1. the currents in rods ba, ad, bc and cd;
2. the external torque needed to maintain uniform rotation.



(a)

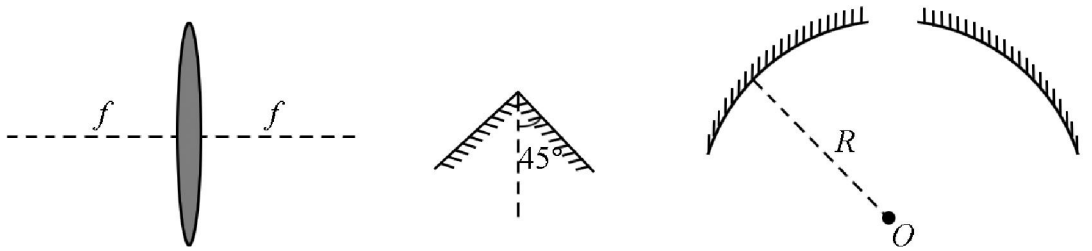


(b)

VIII. (10 points) A hollow laser beam is cylindrical with zero intensity on its axis. It traps tiny particles at the center, acting as optical guides, tweezers or spanners for precise control of nanoparticles and biological cells. Widely used in biology, laser processing and atomic cooling, it is becoming a new subdiscipline.

Optical systems turn solid cylindrical beams into hollow beams. Use a convex lens of focal length f, a mirror with cone angle 45°, and a spherical mirror of radius R with a central circular hole, as shown. Use them to design an optical system converting a very thin solid cylindrical input beam into a hollow output beam with a cylindrical empty core of radius r. Answer:

1. Draw the ray diagram.
2. Find distance x from the cone apex to the spherical mirror's center of curvature.



(a) (b) (c)